Project Overview

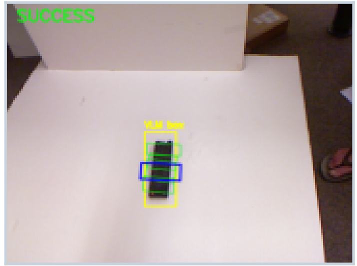
Whole-project progress

RESEARCH QUESTION

Can a pretrained open-vocabulary VLM improve 2D grasp rectangle detection as a localisation front end?

Core idea: keep the localisation front end fixed and compare geometric and learned grasp back ends.

DATASET	BOX COVERAGE	OVERLAP	ORIENTATION
885 samples	**885 / 885**	**IoU $\geq$ 0.25**	**Error $\leq$ 30°**



Example localisation and grasp

Cornell rectangle metric: a prediction is correct if it matches any positive annotation on both criteria (Jiang et al., 2011; Lenz et al., 2015).

Three Experimental Pipelines

1 Traditional CV

Full RGB image
↓
Thresholding + contours
↓
Min-area rotated rectangle

2 VLM + geometry

Grounding DINO box
↓
Segment within the box
↓
Geometric grasp rectangle

3 VLM + CNN

Grounding DINO box
↓
224 × 224 RGB crop
↓
Regress 6 grasp parameters

Experimental Results

Results first

Full Cornell dataset (885 samples)

Method	Success rate	Mean best IoU	Mean angle error
Traditional CV	56.95%	0.3360	29.62°
VLM + geometry	73.33%	0.4182	**14.81°**
VLM + CNN (5 runs)	**74.51% ± 1.38%**	**0.4510 ± 0.0081**	16.49° ± 0.72°

Unseen-object test set (85 samples)

Failure analysis

126 / 236 failures have a correct angle; the main error is position or size.

→ This motivates a learned grasp back end.

Key Findings

+16.38 percentage points

Largest gain from VLM

56.95% → 73.33%

Higher IoU & generalisation

CNN improves geometry

Test: 82.35% ± 4.53%

14.81°

Geometry is best for angle

Explicit priors remain useful

Current Status and Next Steps

Experiments complete; dissertation next

Completed

✓ **Three experimental pipelines**
Traditional CV, VLM + geometry, VLM + CNN

✓ **Full evaluation**
885 Cornell samples with one rectangle metric

✓ **Stability analysis**
Five CNN runs with different random seeds

✓ **Failure analysis**
236 failures categorised by IoU and angle

Next Steps

1. **Write the dissertation**
   Introduction, Methodology, Results, Discussion
2. **Prepare result figures**
   Method comparison, failure modes, examples
3. **Add per-sample analysis**
   Focus on CNN position, size and angle errors
4. **Optional hybrid back end**
   CNN for geometry; explicit prior for orientation

Current Limitations

- Offline 2D evaluation on Cornell only
- No physical robot control or closed-loop grasping
- 885/885 means a box was returned, not that every box was perfect

Key Literature

1. Jiang et al. (2011): grasp rectangles and evaluation
2. Lenz et al. (2015): 30° + 25% IoU success criterion
3. Liu et al. (2023): Grounding DINO open-set detection
4. Morrison et al. (2018): GG-CNN and double-angle encoding

Stage conclusion: VLM localisation is effective; the next focus is grasp back-end refinement and dissertation writing.